

☐ LDEM_80S_80MPP_ADJ.tiff

Verified

Digital elevation model · lunar south pole · terrain layer input

PROVENANCE

Publisher

NASA · Goddard PGDA

Instrument

LOLA (Lunar Orbiter Laser Altimeter)

Original source link

pgda.gsfc.nasa.gov/products/78

Accessed on

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File size (original)

412 MB

TECHNICAL SPECIFICATION

Coverage

80°S latitude to pole

Native resolution

80 m / pixel

Coordinate system

stere, lat_0=-90, R=1737400

Format

GeoTIFF, 32-bit float

Elevation range (raw)

-5,420 m to 3,180 m

No-data pixels

1.8% of grid

PROCESSING PIPELINE

- 1 Reprojected from source CRS to Moon south polar stereographic.
- 2 Cropped to 400 × 400 km analysis grid centred on the pole.
- 3 Resampled from 80 m to 1 km/pixel for cross-layer alignment.
- 4 Derived slope, roughness and flatness layers from this elevation grid.

VALIDATION CHECK

- ☐ Cross-checked crater rim elevations against the Robbins lunar crater database; values matched within expected tolerance.

KNOWN LIMITATIONS

- ☐ Features smaller than 80 m are not resolvable. Some permanently shadowed craters have data gaps due to limited laser returns.